

Noise Analysis — 4um

PAGE 1 — FILE STATISTICS

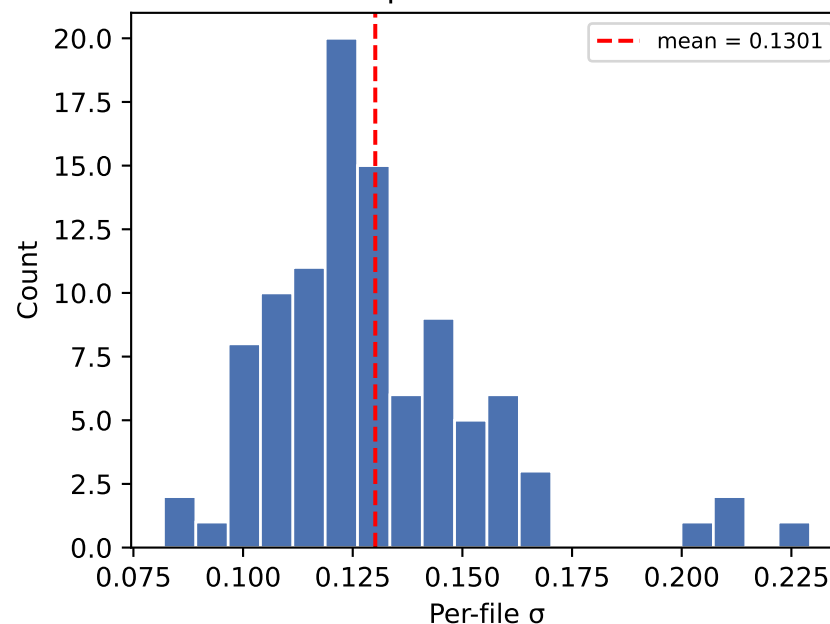
Folder: data/dataset_hamming_clean/test/4um
Files: 100
Signal length: 2500–2500 samples
Dtype: float64

Aggregate noise statistics:

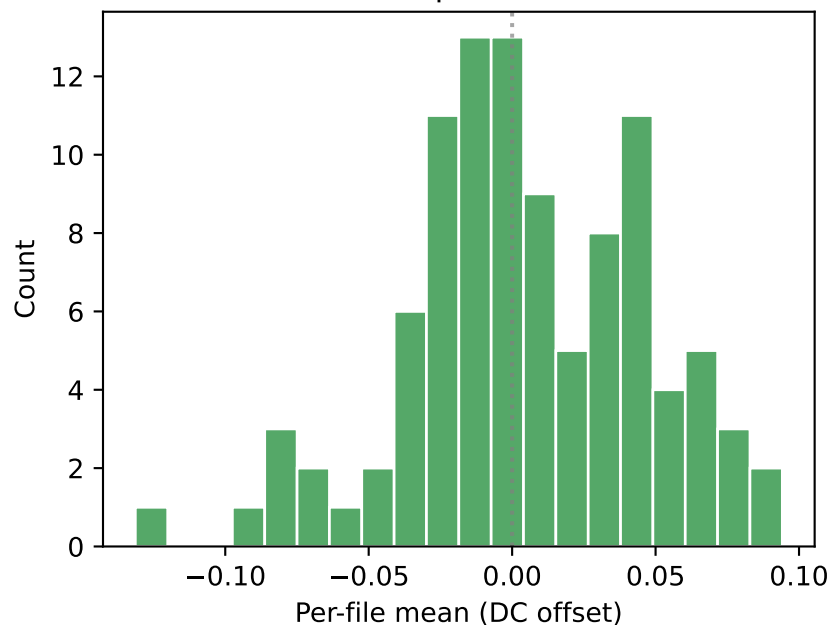
mean(σ): 0.130140
std(σ): 0.024900
CV(σ): 0.1913
mean(RMS): 0.136698
mean(kurtosis): 1.7459 (σ = Gaussian)

mean(DC): 0.004693
std(DC): 0.042075
Amplitude range: [-0.7933, 0.8274]

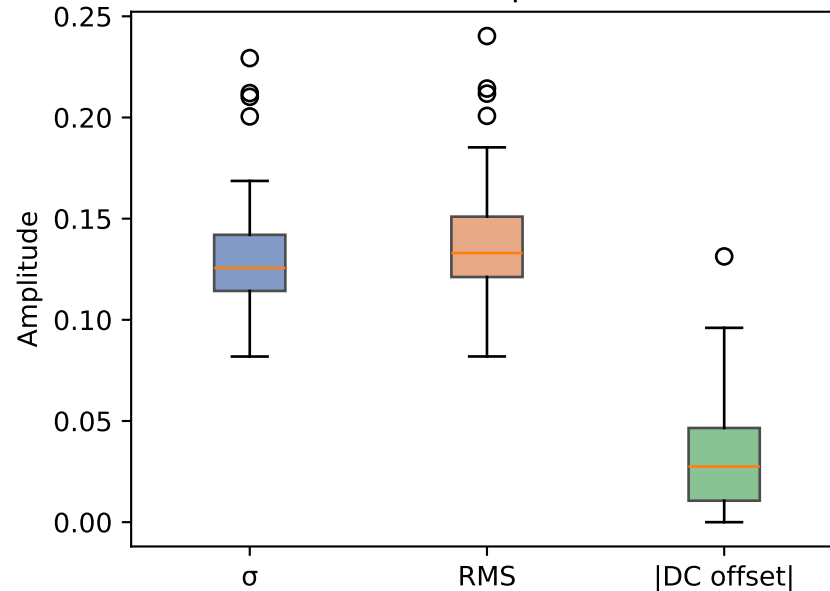
Distribution of per-file standard deviations



Distribution of per-file DC offsets

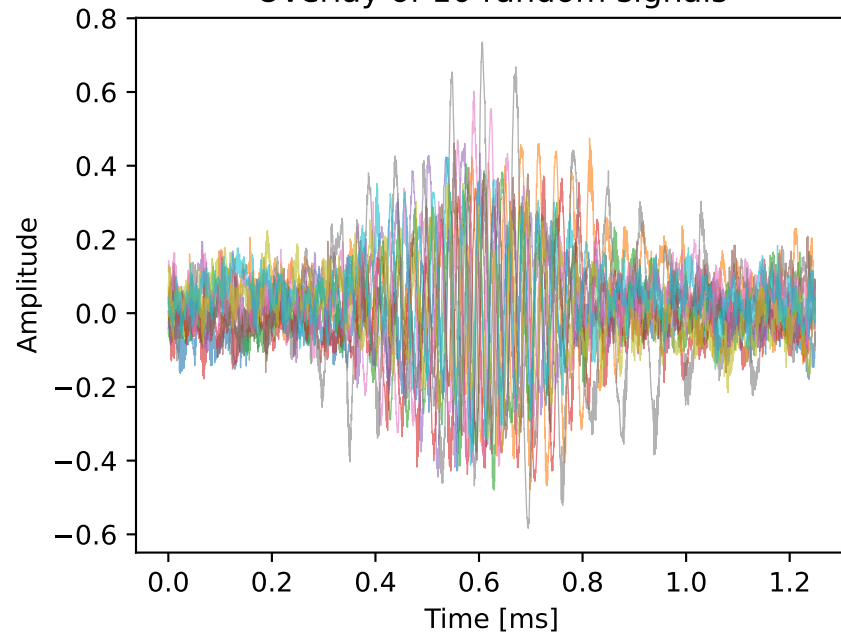


Distribution of amplitude metrics

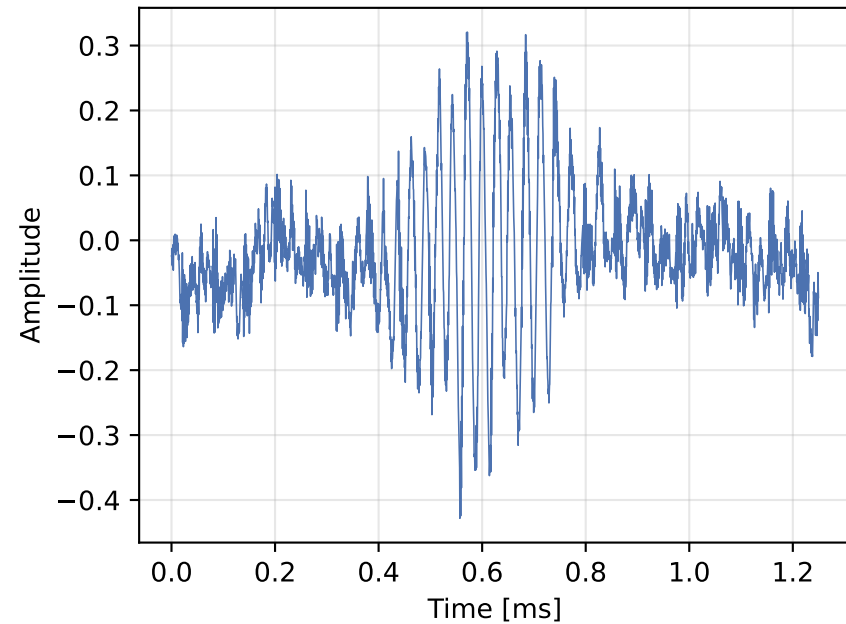


Time-Domain Analysis — 4um

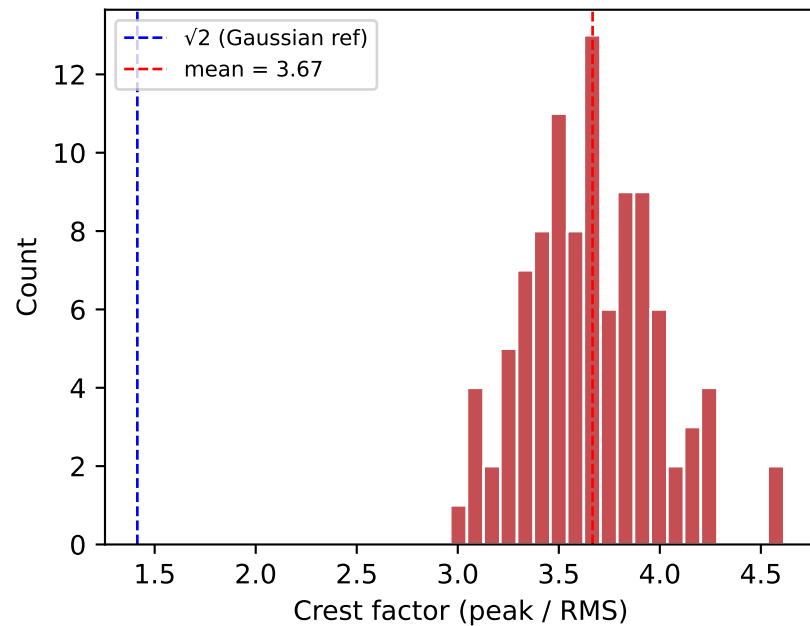
Overlay of 10 random signals



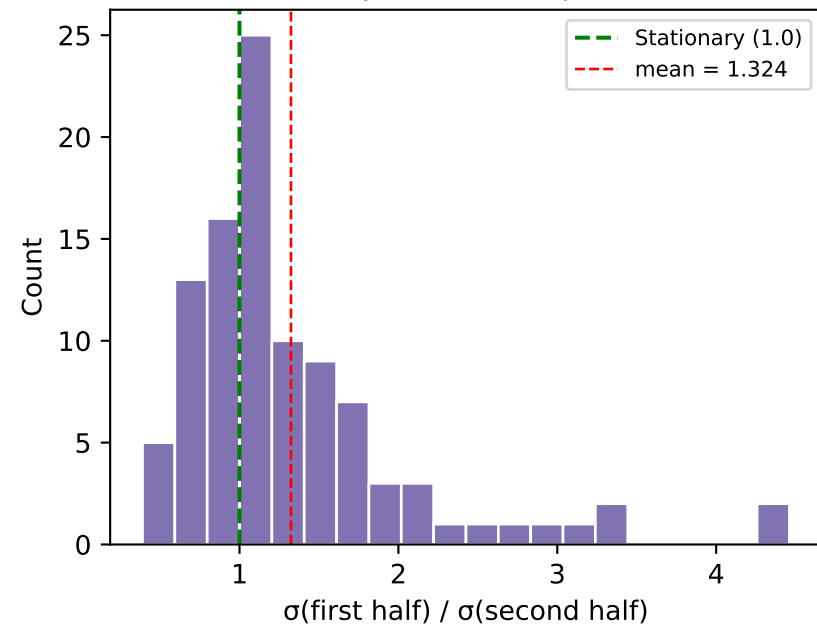
Signal #96 (detail)



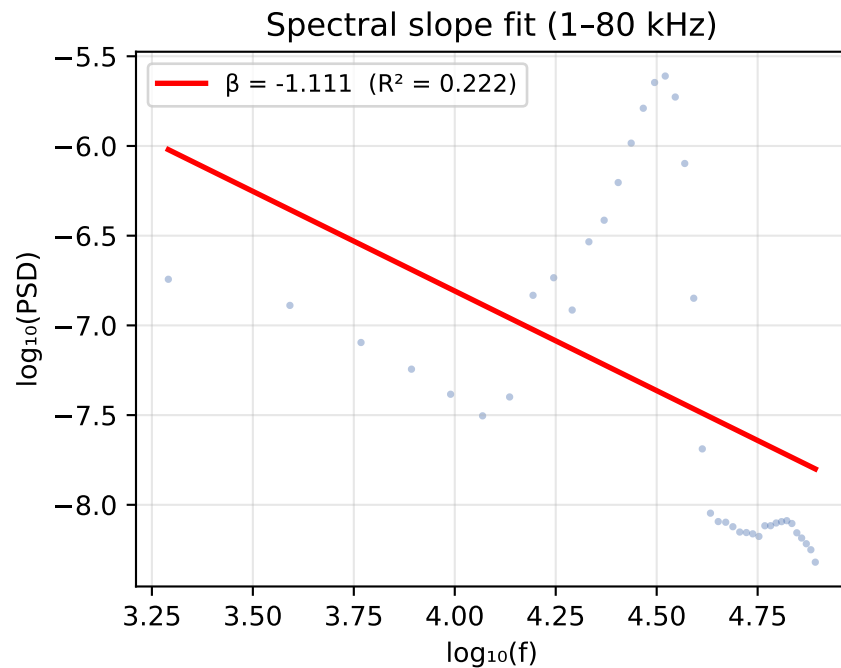
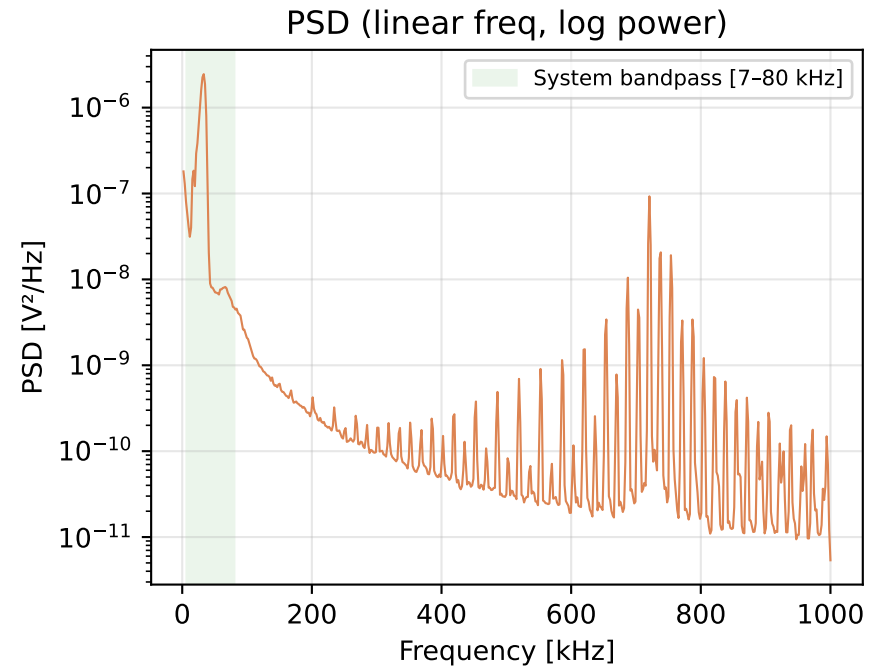
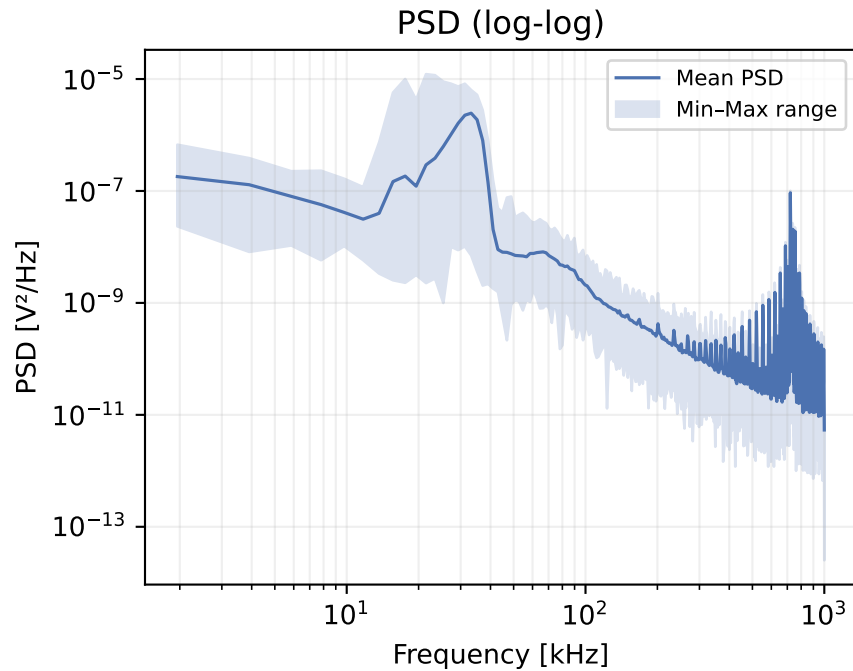
Crest factor distribution



Stationarity test (half-split σ ratio)



Power Spectral Density — 4um



SPECTRAL ANALYSIS SUMMARY

Spectral slope β : -1.1108
Fit quality R^2 : 0.2215
Spectral flatness: 0.0064 (1.0 = white)

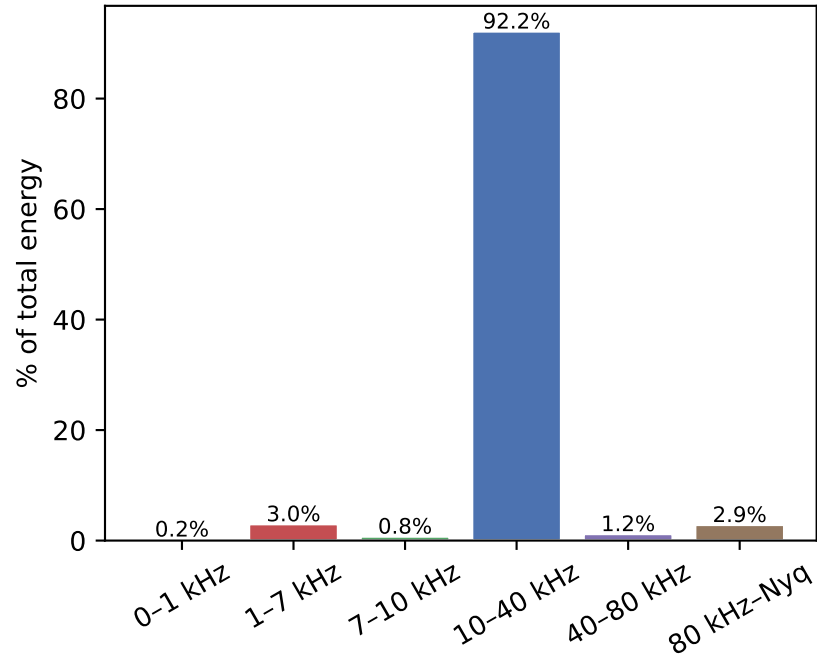
Slope interpretation:

$\beta \approx 0 \rightarrow$ White noise (flat PSD)
 $\beta \approx -1 \rightarrow$ Pink / $1/f$ (equal energy per octave)
 $\beta \approx -2 \rightarrow$ Brown / $1/f^2$ (random walk)

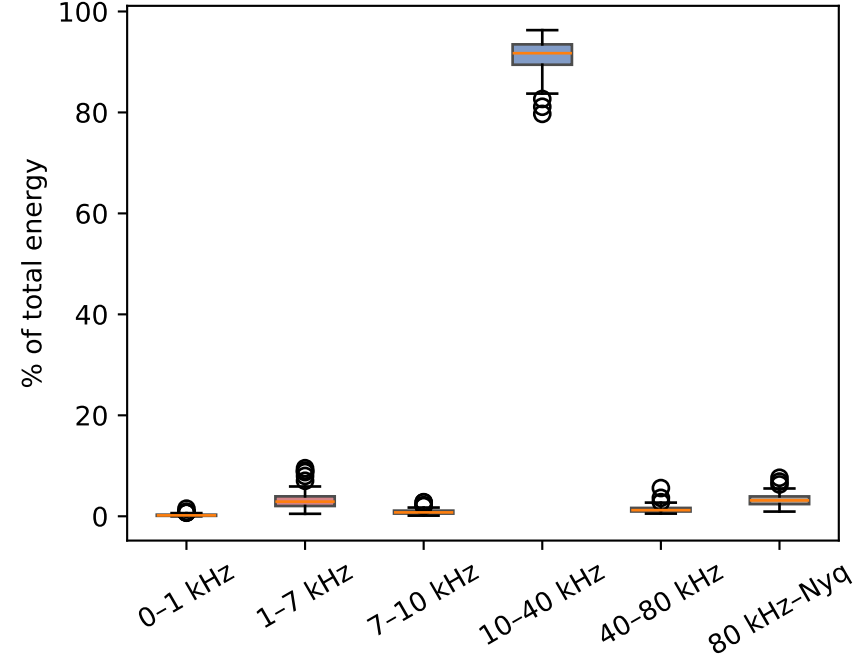
Classification: Pink ($1/f$)

Frequency Band Energy — 4um

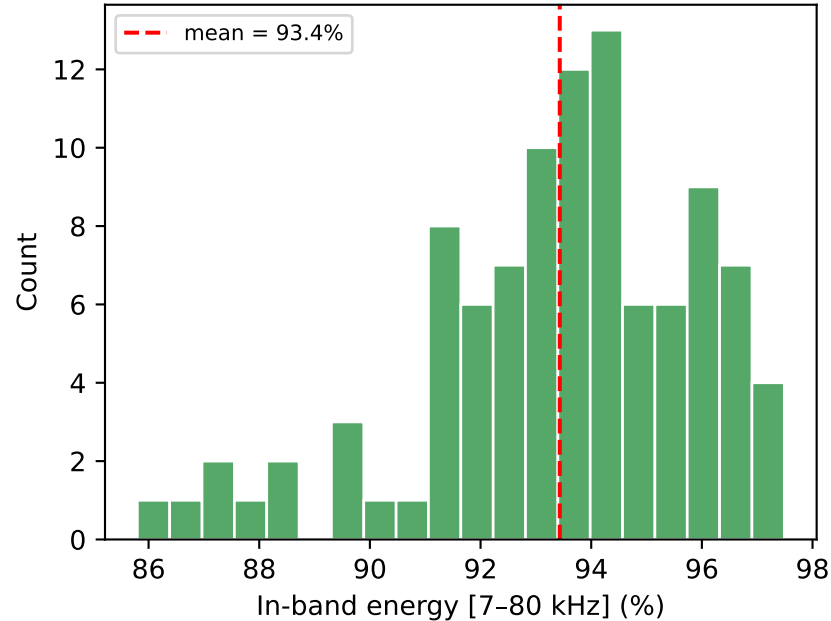
Average energy per frequency band



Per-file band energy distribution



System bandpass energy ratio per file



BAND ENERGY SUMMARY

Band	Mean %	Std %
0-1 kHz	0.2%	0.2%
1-7 kHz	3.0%	1.8%
7-10 kHz	0.8%	0.5%
10-40 kHz	92.2%	3.3%
40-80 kHz	1.2%	0.7%
80 kHz-Nyq	2.9%	1.2%
In-band [7-80k]	93.4%	2.5%

Noise Type Classification — 4um

NOISE TYPE CLASSIFICATION

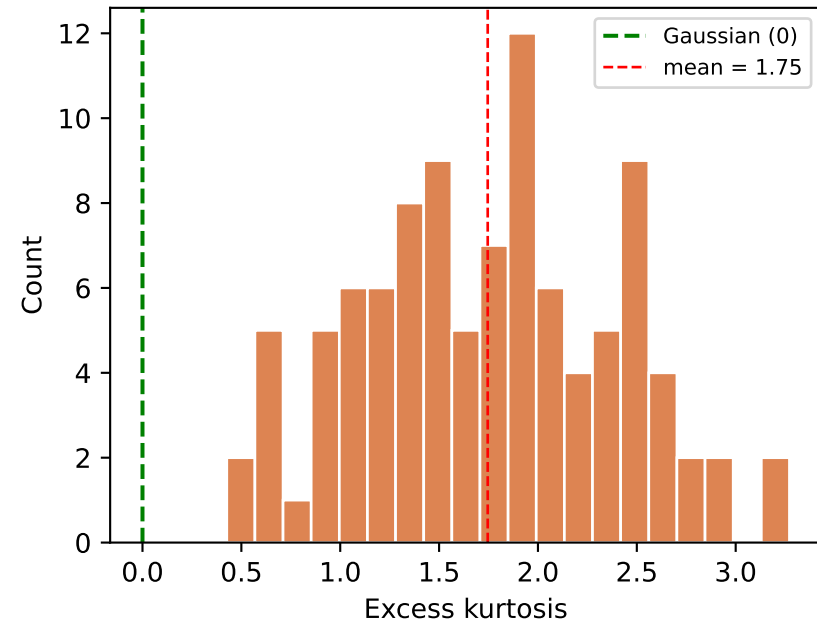
Spectral color: Pink (1/f)
slope β = -1.1108
flatness = 0.0064

Amplitude dist.: Near-Gaussian
kurtosis (mean): 1.7459
(excess kurtosis: 0 = Gaussian,
>0 = heavy-tailed, <0 = light-tailed)

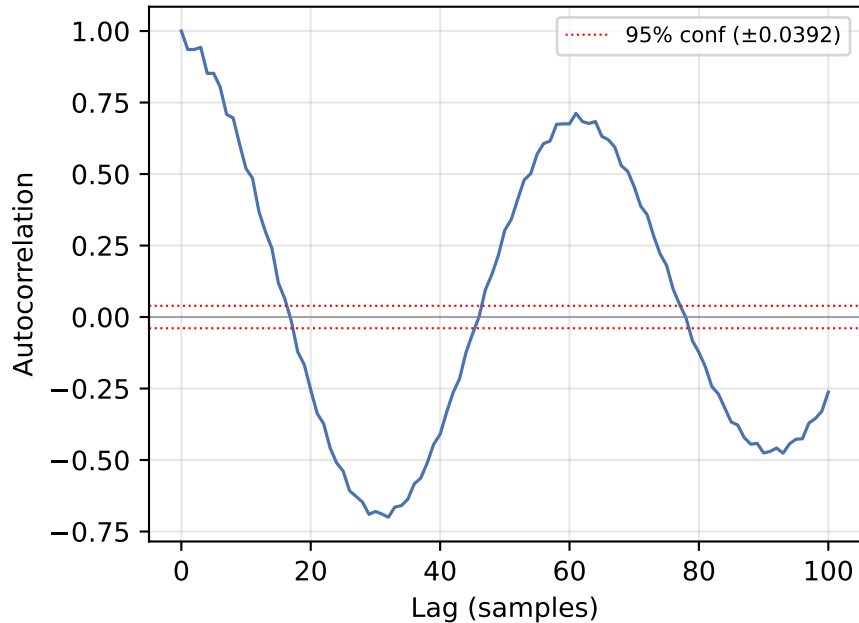
Decision boundaries:
 $|\beta| < 0.3$ → White
0.3–1.5 → Pink (1/f)
> 1.5 → Brown (1/f²)

 $|\text{kurt}| < 0.5$ → Gaussian
0.5–2.0 → Near-Gaussian
> 2.0 → Non-Gaussian

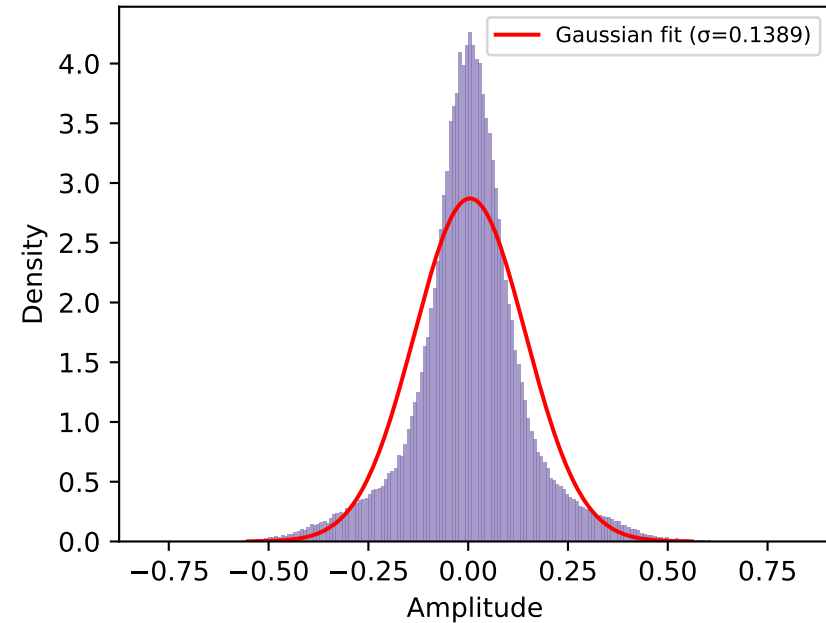
Per-file kurtosis distribution



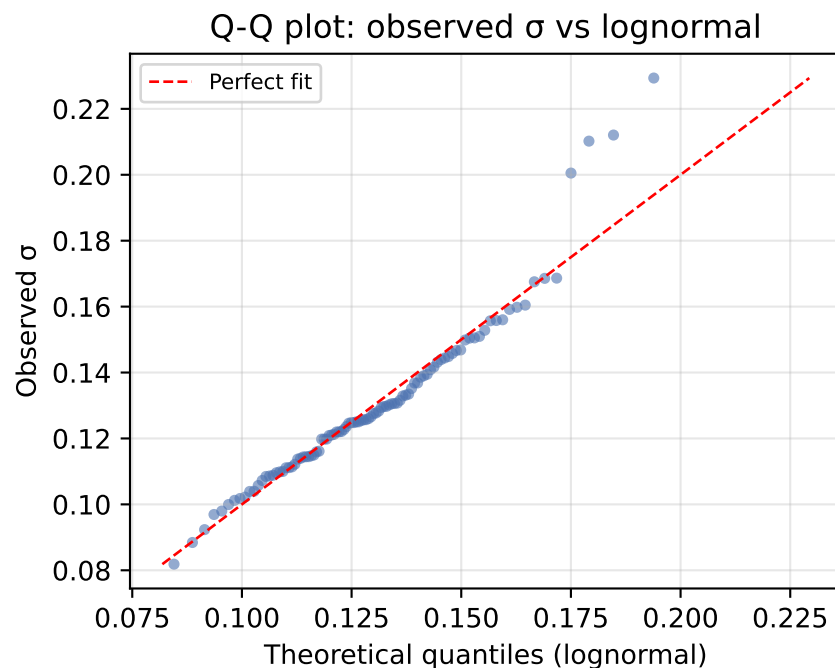
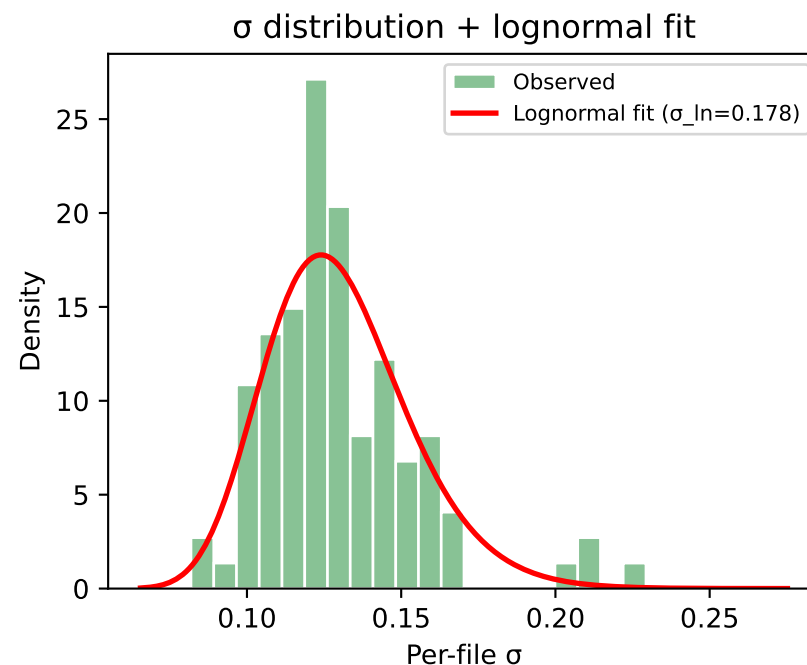
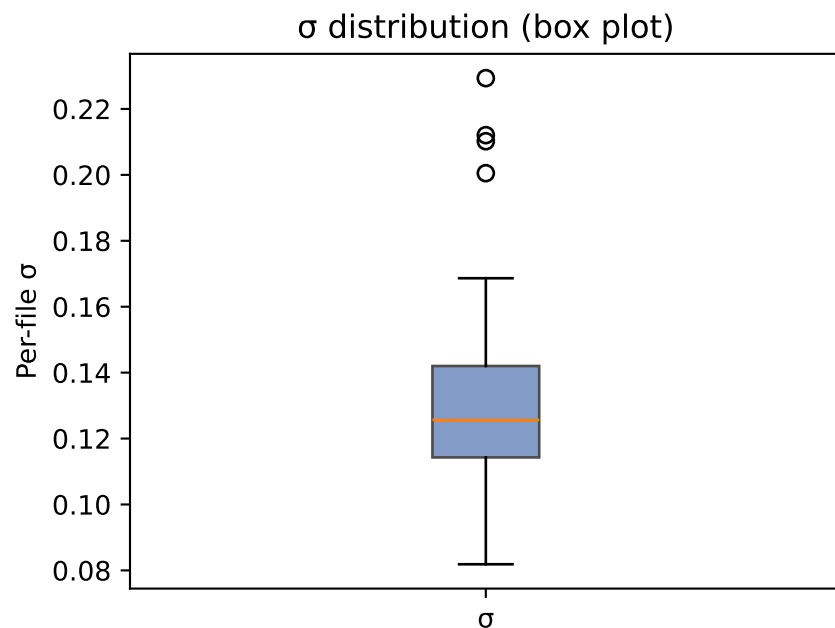
Average autocorrelation (first 100 lags)



Amplitude distribution (all samples pooled)



Amplitude Variability (Inter-file) — 4um



AMPLITUDE VARIABILITY SUMMARY

Number of files: 100
mean(σ): 0.130140
std(σ): 0.024900
CV = std(σ)/mean(σ): 0.1913

Lognormal fit parameters:
shape ($\sigma_{\ln}$): 0.1782
scale ($\exp(\mu_{\ln})$): 0.128009

Reference (from real noise files):
CV_ref = 0.19
Ratio CV/CV_ref: 1.01

Interpretation:
Consistent with real hardware noise (CV $\approx$ 0.19)